INCH-POUND

MIL-DTL-32555A
8 August 2018
SUPERSEDING
MIL-DTL-32555
1 August 2016

DETAIL SPECIFICATION

TOOL, INSTALLATION, FOR ADJUSTABLE TIE-DOWN STRAPS, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

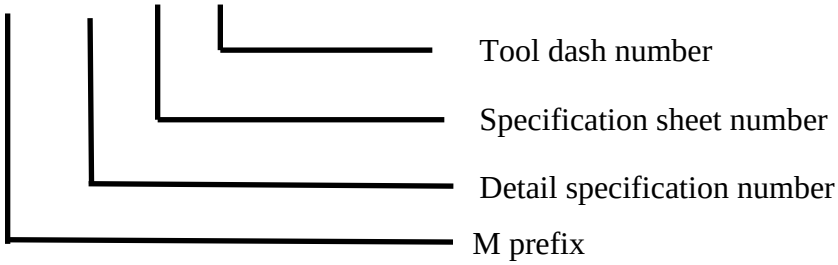
1. SCOPE

1.1 Scope. This specification covers the tool used to install MIL-DTL-32554 tie-down straps on wire bundles (see 6.1).

1.2 Part or Identifying Number (PIN). The PINs to be used for tie-down strap tools acquired to this specification are created as follows:

Example:

M 32555 /01 -01



Comments, suggestions, or questions on this document should be addressed to the Naval Air Warfare Center Aircraft Division Lakehurst, Code 4.1.2.2, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTION

A-A-52084	-	Tape, Lacing and Tying, Aramid

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32554	-	Straps, Tie-Down, Adjustable, Non-Metallic, Cable Bundling, General Specification for
MIL-DTL-32555/1	-	Tool, Installation, for Adjustable Tie-Down Straps

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-202	-	Electronic and Electrical Component Parts
MIL-STD-202-101	-	Method 101, Salt Atmosphere (Corrosion)
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SAE INTERNATIONAL

SAE AIR1351	-	Manufacturers' Identification of Aerospace Electrical and Electronic Wiring Devices and Accessories

(Copies of these documents are available online from www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. The installation tools furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.4).

3.3 Materials and components. Materials and components used in the fabrication of tools covered by this specification shall be capable of meeting the requirements of this specification. The installation tool, including all parts, shall be so constructed and finished that it shall be free from all defects that may affect proper functioning in service. The tool shall have no burrs, sharp edges or sharp corners.

3.3.1 Finish. All metal parts shall withstand the corrosion test of 4.6.6.

3.4 Design and construction. The tool shall be hand operated and capable of applying tie-down straps to their respective wire bundles. The tools shall consist of all components required to attach the strap assembly to the bundle and sever the scrap end of the strap. The tool shall be constructed to withstand stress, shock, vibration and any other conditions incident to service use. The design shall provide for rigidity of parts and resistance to fatigue.

3.4.1 Dimensions. The tool dimensions shall meet the maximum dimensions specified in the specification sheet.

3.4.2 Frame. The frame shall be capable of withstanding the forces encountered during tightening and cutting of the tie-down strap.

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3.4.3 Tool operation. Tool operation shall include tensioning and activating tie-down strap, and cutting excess tie-down strap.

3.4.3.1 Tie-down strap tensioning. The tool shall be designed to apply tension on the free end of the tie-down strap lacing tape (strap). To install the tie-down strap, the free end of the strap is inserted into the rotating tensioning device and pulled until the fastener touches and rests against the tip of the tool. The tool shall apply tension to the MIL-DTL-32554 tie-down strap when the handle is depressed until the required tension to activate the tie-down strap is reached.

3.4.3.2 Cutting excess tie-down strap. Continued depressing of the tool handle after tensioning and activation of tie-down strap, the tool shall cut the remaining tail of the tie-down strap and return to its original position freely as the handle is released. (See 4.6.2). Provisions within the tool shall be made to retain the cut end of the tie-down strap.

3.4.4 Guidance. Provision shall be made to guide the strap into the working position in the tool.

3.4.5 Spring return of handle. The handle of the tool shall return freely and immediately to the fully opened position when released after the tension and cutting operation has been completed.

3.4.6 Compression force.

3.4.6.1 High compression. The tool shall perform satisfactorily and meet the normal compression force requirement after applying a high compression force of 150 pounds. (See 4.6.3).

3.4.6.2 Normal compression. The compression force required to advance the lace tape of the tie-down strap, apply required tension and cut-off the tie-down strap shall be 30 pounds-force (lbf) maximum and shall result in 17 to 22 lbf tensile force applied to the free end of the lace tape to activate the tie down fastener (see 4.6.4).

3.5 Performance. The tools shall comply with all the test and inspection requirements of this specification.

3.5.1 Low temperature. There shall be no binding of the tool handles and the gripping and cutting devices, when tested as specified in 4.6.5.

3.5.2 Corrosion. After the corrosion test of 4.6.6, the tool shall comply with the low temperature performance requirements of 3.5.1. There shall be no damage to the tool that impairs operation nor evidence of corrosion, when tested as specified in 4.6.6.

3.5.3 Life cycle. The tool shall withstand a total of 15,000 tension and cut cycles in accordance with 4.6.7.

3.6 Workmanship. The installation tool, including all parts, shall be so constructed and

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finished that it shall be free from all defects that may affect proper functioning in service. There shall be no sharp edges, burrs, or other defects that are potential hazards to the operator of the tool.

3.7 Instructions. If specified in the contract (see 6.3.d), each tool shall be supplied with written instructions which include, as a minimum, gaging and calibration (if required) methods, a list of replacement part numbers, and safe operation and tool storage instructions.

3.8 Marking. All marking shall be in accordance with MIL-STD-130 and shall be permanently legible using either nameplates, dataplates, or direct marking.

3.8.1 Supplier's (manufacturer's) identification. The supplier's name and symbol, or only the symbol shall be marked in the area specified on the specification sheet. The symbol alone shall be used when the available space does not allow for legible marking of the manufacturer's name. If a symbol is used, it shall be a symbol listed in SAE AIR1351.

3.8.2 Nameplates and dataplates. Nameplates and dataplates on tools shall be marked with information in accordance with the applicable specification sheet. The material of the plates and the mounting of the plates shall be capable of withstanding the environmental tests specified for the item. Marking shall be permanent to the extent required for utilization of the item.

3.8.3 Direct markings. Tools, dies, and accessories shall be directly marked with information in accordance with the applicable specification sheet. Part numbers shall be permanently marked on the dies or blades.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein, are classified as follows:

a. Qualification inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at ambient temperature and humidity as specified in MIL-STD-202.

4.2.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy and quantity to permit performance of the required inspection shall be established and maintained by the contractor.

4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests in table I. Unless otherwise specified in table I, sequential testing is required. The qualifying activity shall perform or witness the particular tests as specified in table I and the supplier shall perform the remaining tests.

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4.3.1 Qualification inspection process. For guidance on the qualification inspection process, see 6.4 and 6.4.1.

4.3.2 Qualification inspection samples. The qualification test samples shall consist of three (3) complete tool assemblies.

a. Qualifying activity tests. For the qualifying activity tests, 4 complete tool assemblies with one hundred MIL-DTL-32554 tie-down straps and 32,000 feet of A-A-52084-C-2 lacing tape (only required for life cycle testing) shall be separately packaged and forwarded to the address (see 6.4) designated in the letter of authorization. Each sample shall be identified by marking each package with the following information:

Sample for qualification:

Specification MIL-DTL-32555, Tool, Installation, for Adjustable Tie-down Straps
Part number M32555/XX-XX
Name of supplier
Supplier's part number and Commercial and Government Entity (CAGE) Code
Submitted (date) under authorization (reference letter authorizing the inspection)

4.3.3 Retention of qualification. Retention of qualification inspections shall be performed every 36 months after the original qualification approval date. The qualifying activity shall perform or witness the tests specified in table II. Unless otherwise specified in table II, sequential testing is required. The qualifying activity may shift the reporting date to accommodate testing schedules, but no later than 18 months from the retention of qualification due date.

4.3.4 No production certification. If no production of the qualified products has occurred for the entire reporting period, the supplier shall provide certification to the qualifying activity that no changes in the product materials, design, manufacturing processes, or site of production have occurred since qualification approval (see 4.3). Certification for more than one reporting period is not permitted. The supplier shall contact the qualifying activity to determine the conditions for certification. The supplier shall submit the certification on DD Form 1718 provided by the qualifying activity.

4.3.5 Qualification of additional tools. Qualification by similarity to qualified components or to components submitted for qualification is permissible when materials, designs, and manufacturing processes are similar. The manufacturer shall provide to the qualifying activity, full details of the similarities and differences of the components and manufacturing processes, and a proposed qualification test program that will address the differences. Testing shall not begin until the manufacturer's proposal has been approved or modified by the qualifying activity.

4.4 Conformance inspection. Conformance inspections shall consist of:

a. End item inspection (see 4.4.1).
b. Sampling inspection (see 4.4.2).

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4.4.1 End item inspection. Each tool shall be subjected to the examination of product (see 4.6.1). Alternately, a system of in-process controls accepted by the qualifying activity is permitted to be substituted for individual end item inspection in accordance with MIL-STD-1916.

4.4.2 Sampling inspection. Samples shall be selected at random in accordance with MIL-STD-1916 for lot sampling by attributes, verification level 1. There shall be no damage to the tools that are selected when tested in accordance with the following requirements:

a. Spring return of handle (see 4.6.2).
b. High compression force (see 4.6.3).
c. Normal compression force (4.6.4).

TABLE I. Qualification inspections.

Examination or Test	Requirement Paragraph	Test Paragraph (in sequence)
Examination of Product 1/, 2/	3.4, 3.6, 3.7, 3.8	4.6.1
Spring return of handle	3.4.5	4.6.2
High compression force	3.4.6.1	4.6.3
Normal compression force	3.4.6.2	4.6.4
Low temperature performance	3.5.1	4.6.5
Corrosion 2/, 3/, 4/	3.5.2	4.6.6
Life cycle 2/, 3/ 4/	3.5.3	4.6.7

1/ Conformance inspection test.
2/ Tests conducted by or witnessed by the Qualifying activity.
3/ Use one tool for life testing and a second tool for corrosion testing.
4/ Required only for initial qualification.

TABLE II. Retention of qualification inspections.

Examination or Test	Requirement Paragraph	Test Paragraph (in sequence)
Examination of Product	3.4, 3.6, 3.7, 3.8	4.6.1
Corrosion	3.5.2	4.6.6

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4.5 Test conditions.

4.5.1 Atmospheric conditions. Unless otherwise specified herein, all tests shall be conducted at an atmospheric pressure of 28 to 31 inches of mercury, a temperature of 68 ºF to 85 ºF (20 ºC to 29.44 ºC) and a relative humidity of 30 to 80 percent.

4.6 Inspection methods.

4.6.1 Examination of product. Installation tools and their component parts shall be visually examined to determine conformance to the requirements of this specification.

4.6.2 Spring return of handle. This test shall be conducted with and without each adjustable tie-down strap accommodated by the tool. The handle and gripping device assembly of the tool shall automatically return to their fully open position upon completion of the tension and cutting operations. The tool shall be tested in the vertical and horizontal planes. When testing in the horizontal plane, the gripping device shall be facing downward (see 3.4.5).

4.6.3 High compression force. The tool shall perform satisfactorily after a compression force of 150 pounds is exerted on the tool handle at a point 1 ±1/8 inch from the extremity of the handle (see 3.4.6.1).

4.6.4 Normal compression force. The maximum normal compression force of 30 pounds shall not be exceeded when installing (tensioning and cutting off operation) the tie-down cable strap. The compression force shall be measured at a point on the tool handle 1 ±1/8 inch from the extremity of the handle (see 3.4.6.2).

4.6.5 Low temperature performance. The tie-down strap tool, four of the tie-down straps, and wire bundle, fabricated in accordance with MIL-DTL-32554, shall be maintained at a temperature of 5 °F ±3.6 °F (-15 ºC ±2 ºC) for a period of one hour. The tie-down strap shall then be installed to its wire bundle while at this temperature. The tool shall continue to function during and after this test. The test straps shall comply with the requirements of MIL-DTL-32554. It is permissible for a change in the tightness of the bundle if the temperature of the bundle is allowed to return to normal environmental conditions (see 3.5.1).

4.6.6 Corrosion. One tool shall be subjected to a salt spray (corrosion) test in accordance with MIL-STD-202-101, Test Condition A. The tool shall then be dried. After the tool is dry, it shall comply with 4.6.6. If the tool is metallic, it shall not exhibit base metal on plated or treated parts. White or black residue is permitted as long as the tool function is not impaired (see 3.5.2).

4.6.7 Life cycle. The tool shall withstand a total of 15,000 tension and cut cycles. Aramid lacing tape conforming to A-A-52084-C-2 shall be used. The lacing tape material shall be fed to the tool in such a manner that the action of the gripping device, when the tool is operated, shall exert tension corresponding to the tensile force required for activation of the fastener head and cutting of the lace material of the tie-down strap without damage to the assembly. A cycle shall consist of closing the tool through the cutting position and returning the handle to the full open

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position. At no time during this test shall the compression force, required to be exerted on the handles of the tool at a point 1 ±1/8-inch from the extremity of the handles, exceed the normal compression force specified in 4.6.4. After each 2,500 cycles, the tool shall be tested by installing three MIL-DTL-32554 tie-down straps around a bundle of wires fabricated in accordance with MIL-DTL-32554. The cutting blades may be replaced after each 2,500 cycles (see 3.5.3).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract of order (see 6.3). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or with the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Hand-operated tools covered by this specification are intended to be used for installation of MIL-DTL-32554 tie-down straps.

6.2 Application note. The tool may be used for single and multiple wrapping of MIL-DTL-32554 tie-straps.

6.3 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of the specification.
b. Tie-down strap tool PIN (see 1.2).
c. Quantity desired.
d. Instructions (see 3.7).
e. Special markings, if required (see 3.8).
f. Packaging requirements (see 5.1).

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List (QPL) No. 32555 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the qualifying activity, Naval Air Systems Command (Code 4.4.5.3), 48298 Shaw Road,

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Building 1461, Patuxent River, MD 20670-1900 or QPL_Support@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4.1 Qualification process. To initiate the qualification process, prior to testing, prospective suppliers must forward a written request for qualification to the qualifying activity (see 6.4). The qualifying activity will respond in writing with a formal "letter of authorization" providing detailed instructions for the submission of product samples. Testing cannot begin until the supplier has received this authorization letter. The qualification process is further described in the SD-6 "Provisions Governing Qualification." Copies of this document are available online at https://assist.dla.mil.

6.4.2 QPL supplemental information. The qualifying activity provides a summarized list of all qualified sources on a publicly accessible website http://www.navair.navy.mil/qpl/. The summary includes but is not limited to the supplier approved part number and related specification part number, a dedicated approval reference number, a supplier location where purchases may be requested, and the manufacturing location of the component.

6.5 Subject (key word) listing.

Frame
Gripping device
Hand operated

6.6 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS
Project 5120-2018-005

Review activities:
Army - AV, MI
Navy - EC, OS, SH
Air Force - 11, 20
DLA - GS

Industry association:
SAE AE-8C2

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